

Retrieval-Augmented Generation for Smart Classroom Applications

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Abstract—Our study proposes an intelligent teaching method based on RAG, which dynamically integrates authoritative knowledge bases with language models to address the limitations of large models in factual accuracy and domain-specific expertise. The experiments employ three evaluation metrics to validate the effectiveness of RAG in educational scenarios. The results demonstrate that the RAG-enhanced model (54.8% accuracy) significantly outperforms baseline models (Llama2-7B at 38.8%), and ablation studies confirm the critical role of the retrieval module and factual verification in performance improvement. The system supports personalized teaching resource generation, offering a new paradigm for AI-driven education.

Keywords—RAG, Education, Language models

I. INTRODUCTION

Recent breakthroughs in artificial intelligence have been largely fueled by the evolution of deep neural networks, especially powerful language models built on the Transformer framework [1]. Systems like GPT, BERT, and T5 [2] have revolutionized how machines process human language through their ability to learn from massive text corpora. These sophisticated models exhibit human-like comprehension and text production skills, making them versatile tools for applications ranging from creative writing assistance and cross-lingual translation to interactive dialogue systems.

While large language models demonstrate remarkable text generation capabilities, they still grapple with significant limitations regarding factual reliability and domain expertise. A persistent issue known as "hallucination" [3] occurs when these systems generate convincing but factually inaccurate responses - for instance, inventing historical events when answering questions. This stems from their fundamental design: these models operate solely based on their pre-existing training data, lacking the ability to dynamically consult authoritative knowledge sources or verify information in real-time. Consequently, their outputs may contain outdated data or imprecise information, particularly when addressing specialized subjects that demand expert-level knowledge.

To overcome the educational constraints of LLMs, especially their inability to access current information, this research employs Retrieval-Augmented Generation (RAG) - an

innovative approach that merges dynamic knowledge retrieval with advanced text generation [4]. The proposed solution establishes a synergistic pipeline where educational databases continuously feed updated information to language models, enabling the creation of precise, pedagogically optimized content. This integration significantly improves three key educational aspects: (1) instructional delivery efficiency, (2) adaptive learning personalization, and (3) intelligent educational support [5]. The RAG methodology implements a two-phase architecture: initially, a specialized retrieval component dynamically gathers authoritative content from curated sources (including curriculum materials and academic publications), which subsequently undergoes contextual enhancement through generative AI before final output.

The integration of RAG technology in education opens new possibilities for enhancing instructional design and student support [6]. Specifically, this framework enables the automated creation of curriculum-aligned teaching resources, real-time response generation for learner inquiries, and the development of individualized study materials. Beyond these applications, the system can intelligently produce targeted practice activities adjusted to each student's competency level, thereby promoting truly personalized education [7]. This study investigates how such AI-augmented approaches can transform traditional pedagogy, with the broader goal of advancing both the effectiveness and accessibility of learning experiences.

This investigation will center on "Retrieval-Augmented Generation for Smart Classroom Applications" as its primary research focus. The methodology involves three key phases: (1) developing a structured pedagogical knowledge repository [8], (2) refining the RAG architecture for educational applications, and (3) conducting empirical validation in active learning environments. Through rigorous assessment of instructional efficacy and learning gains, the project simultaneously examines the technology's capacity to bridge educational disparities, ultimately contributing to next-generation AI-driven education solutions [9].

II. SETTING OF EVALUATION INDICATORS

A. Special tokens

TABLE I. CRITERIA FOR EVALUATION

Type	Output	Definition
Retrieve	{Yes, No}	Whether retrieval is required
Accuracy	{Correct, Partially Correct, Incorrect }	Is the response content accurate and error-free?
Knowledge Coverage	{Fully Covered, Mostly Covered, Partially Covered, Little Covered, No Covered }	Does the retrieved knowledge comprehensively cover all aspects of the question?

This research utilizes three assessment criteria as shown in Table 1. The primary metric, "Knowledge Retrieval Trigger," evaluates whether subsequent LLM output necessitates supplementary information fetching, with binary outcomes: Trigger (external retrieval needed) or Bypass (direct generation sufficient). The decision is made through contextual analysis of both current input and prior outputs (when applicable), assessing the need for factual reinforcement. "Bypass" applies when content is either non-factual or already optimal without retrieval, while "Trigger" activates when accuracy or comprehensiveness requires enhancement through external knowledge. Furthermore, we've incorporated a persistent evidence referencing system, enabling ongoing utilization of previously acquired information [10].

The second metric, "Factual Precision," assesses the veracity of response content, particularly for fact-based inquiries, with three classification levels: Verified (fully accurate), Qualified (partially accurate), and Contradicted (inaccurate). Given that retrieved knowledge may contain imperfections, this measure evaluates whether the supporting evidence delivers reliable and correct information (Verified), or includes inaccuracies, deceptive elements, or obsolete content (Contradicted). During retrieval evaluation, Factual Precision serves as a critical assessment criterion, as flawed knowledge could result in generated responses that diverge from factual truth, ultimately compromising the system's output credibility.

The third metric, "Information Completeness," evaluates the breadth and depth of retrieved knowledge in addressing query requirements, with five assessment tiers: Comprehensive (complete coverage), Substantial (significant coverage), Moderate (partial coverage), Limited (minimal coverage), and Absent (no coverage). This criterion examines how thoroughly the obtained information responds to the primary inquiry, supporting elements, and potential contextual nuances [11]. Our analysis distinguishes three coverage levels: Comprehensive denotes all pertinent dimensions are adequately addressed; Moderate signifies only selective aspects are covered; while Absent indicates essential content is missing, yielding an inadequate response. This evaluation parameter is vital for guaranteeing the thoroughness and applicability of system outputs. For instance, when querying "What are the tourist attractions in Beijing?", retrieval results mentioning only the Great Wall while excluding the Forbidden City and other landmarks would be classified as having Limited coverage.

The RAG workflow implemented in this study operates through the following sequence: Initially, the system performs a retrieval necessity assessment. When retrieval is deemed unnecessary, the model proceeds with conventional autoregressive text generation, identical to baseline language model operation. When external knowledge integration is required, the system sequentially executes three evaluation phases: (1) generating a verification token to assess the factual correctness of retrieved content, (2) producing a coverage assessment token to evaluate the comprehensiveness of retrieved information relative to the query, and (3) outputting a quality assurance token to determine the response's overall efficacy. This multi-stage validation process ensures rigorous quality control throughout the knowledge integration pipeline.

The framework employs probabilistic scoring derived from the aforementioned evaluation metrics. For instance, when the [Comprehensive] token appears in model outputs, this reflects the language model's inference-time probability distribution across all possible coverage classifications ([Comprehensive], [Partial], [Insufficient]), with [Comprehensive] being the maximum likelihood selection. Our methodology computes metric-specific scores by normalizing the conditional probabilities of target tokens. Regarding soft-constrained retrieval activation, the system initiates knowledge fetching when meeting the threshold conditions defined in Equation 1. This represents a normalized relevance scoring function that optimizes the trade-off between pertinent and extraneous information probabilities.

$$\frac{p(\text{Retrieve} = \text{Yes})}{p(\text{Retrieve} = \text{Yes}) + p(\text{Retrieve} = \text{No})} > \delta \quad (1)$$

The Accuracy scoring mechanism implements a weighted evaluation scheme as formalized in Equation 2. This metric applies differential weighting coefficients ($\alpha=1.0$ for Verified responses, $\beta=0.5$ for Qualified responses) while explicitly excluding Contradicted instances from the scoring computation. The weighting strategy reflects the relative confidence in different levels of factual alignment, with full credit awarded for completely accurate information and partial credit for responses containing limited inaccuracies. Among them,

$$S = \sum_{t \in \{\text{Fully}, \text{Partially}, \text{No}\}} : \quad (2)$$

$$s(\text{Accuracy}) = \frac{p(\text{Accuracy} = \text{Correct})}{S} + 0.5 \times \frac{p(\text{Accuracy} = \text{Partially})}{S}$$

The Knowledge Coverage (KC) metric employs a weighted scoring framework formalized in Equation 3. This evaluation scheme implements differential weighting factors ($\omega_1=1.0$ for Comprehensive coverage, $\omega_2=0.5$ for Moderate coverage) while systematically excluding Absent cases from the weighted aggregation. The weighting paradigm reflects the progressive degradation in scoring corresponding to decreasing levels of knowledge coverage completeness, among them,

$$S = \sum_{t \in \{\text{Fully}, \text{Partially}, \text{No}\}} : \quad (3)$$

$$s(\text{KC}) = \frac{p(\text{KC} = \text{Fully})}{S} + 0.5 \times \frac{p(\text{KC} = \text{Partially})}{S}$$

III. EXPERIMENT AND ANALYSIS

A. Data Preparation

The raw data is sourced from the classroom recordings on the intelligent course platform of Shanghai Dianji University. The speech content of the teacher and students is recognized through speech recognition, and all conversations of 45 minutes in each class are recorded. Then, the classroom recording is fed into the large model, and fifty pieces of data related to the content of the lesson are refined through prompt words [12]. The markdown format of prompt words is shown below.

Classroom Dialogue Information Extraction Specification

Task Description

Extract ****50** core teaching information points****** from a ****45-minute** classroom dialogue recording******.

Input Data

Source

Timestamped teacher-student dialogue text.

Content Types

- Knowledge point explanations
- Classroom questions
- Student answers
- Key emphasis
- Teaching instructions
- Classroom interactions

Processing Requirements

1. Exclude content-irrelevant dialogues (e.g., classroom discipline management, equipment adjustments).
2. Merge continuous dialogues on the same topic into a single entry.
3. Ensure each information point remains ****independent and complete****.
4. Prioritize extraction when ****important concepts first appear****.
5. Maintain ****question-answer correspondence**** between teachers and students.

Output Format

```
{
  "id": "Unique serial number (1-50)",
  "type": "Information type",
  "speaker": "Speaker identity (teacher/student)",
  "content": "Refined dialogue content",
  "timestamp": "Start time (MM:SS)",
  "keywords": ["Keyword1", "Keyword2"]
}
```

TABLE II. STRUCTURE AFTER DATA PROCESSING

Key	Value
question	String type, pose a specific question
answers	List type, containing one or more strings as answers to a question
output	String type, model output answer, consistent with the content in answers
target_output	String type, expected target output, consistent with output and answers
instruction	String type, guidance or instruction for the question, consistent with or more concise than the content of the question
preceding_sentences	Previous information
q_id	Number type, unique identifier of the problem
sent_idx	Index of numerical types, sentences or questions
evidence	String type, providing background knowledge or reference sources related to the problem (such as Wikipedia excerpts)
dataset_name	String type, name of data source

In the data processing flow, the cleaned text data is first input into the large model, and the necessity of retrieval is detected through preset Retrieve prompt words. The detection process will analyze the text content and automatically add special tags to identify whether further retrieval and supplementary background knowledge are needed.

After completing the tagging, the system will search for relevant paragraphs based on the data from the Chinese version of Wikipedia. The retrieval module will match key information in the problem and return relevant background knowledge. The search results are stored in structured data format, where each element is a dictionary containing multiple fields, representing a question answer pair, as shown in Table 2.

After data processing, this article uses the Deepseek big model to specially label the data, then trains the critic model and generator model, and finally fine tunes Deepseek 7b chat to obtain a new introspection model [13]. As shown in Figure 1, the reason for the score of 4 is explained in raw_output. This is special tokens generated by the large model, indicating that the answer basically meets the query requirements, but there is still some small room for improvement, such as providing more detailed information, clearer structure, or improving coherence.

<pre> { "input": { "instruction": "Given four answer candidates, A, B, C and D, choose the best answer choice.## Input:\n\nHow did the Glorious Revolution and Bill of Rights pave the way for Britain\u2019s Industrial Revolution, per class?\nA: Revolutions were irrelevant.\nB: Glorious Revolution and Bill of Rights laid foundations.\nC: Industrialization was spontaneous.\nD: Laws hindered growth.", "output": "B", "input": "", "id": "MKS_History_SC_000115", "dataset_name": "MKS", "score": 4, "explanation": 4, "raw_output": "Perceived utility: 4\nExplanation: The response correctly identifies the answer choice that aligns with the historical context, suggesting that the Glorious Revolution and Bill of Rights laid the foundations for Britain's Industrial Revolution. This provides a direct and accurate answer to the question, which is helpful and informative. However, the response does not elaborate on why choice B is correct, which could provide additional insight and context, thus justifying a score of 4 rather than a perfect 5." } } </pre>

Fig. 1. Special tokens of Utility

B. Experimental Results

query_1 = "Leave odd one out: twitter, instagram, whatsapp." query_2 = "Can you tell me the difference between llamas and alpacas?" Model prediction: Twitter, Instagram, and WhatsApp are all social media platforms.[No Retrieval]However, WhatsApp is a messaging app, while Twitter and Instagram are both primarily used for sharing photos and videos.[No Retrieval]Therefore, WhatsApp is the odd one out in this group.[Utility:5] Model prediction: Sure! [Retrieval]* Alpaca (left) and llama (right) in the Andes of southern Peru. Alpacas and llamas are both domesticated species of South American camelids.[Continue to Use Evidence]Alpacas are much smaller than llamas, with a shoulder height of 3 to 4 feet.[Continue to Use Evidence]They are also bred specifically for their fiber, which is used to make all sorts of textiles and clothing.

Fig. 2. Model Q&A

As shown in Figure 2, two instructions are proposed. The first one is "Leave odd one out: twitter, instagram, whatsapp." The fine-tuned model's response is: "Twitter, Instagram, and WhatsApp are all social media platforms.[No Retrieval] However, WhatsApp is a messaging app, while Twitter and Instagram are both primarily used for sharing photos and videos.[No Retrieval] Therefore, WhatsApp is the odd one out in this group.[Utility:5]." This indicates that the instruction does not require retrieval and can be answered solely based on the large model's own knowledge [14]. The Utility score of 5 demonstrates that the response fully meets the requirement.

TABLE III. ABLATION EXPERIMENT

Our data (acc)	
Our-RAG	47.4
Training	
No Critic	44.2
Test	
No retrieval	31.2
No accuracy	30.5
No KC	45.1

TABLE IV. BASELINE COMPARISON

Our data (acc)	
LM	
Llama2 _{7B}	38.8
Alpaca _{7B}	48.2
Mistral _{7B}	51.3
Our-RAG _{7B}	54.8

The second instruction is: "Can you tell me the difference between llamas and alpacas?" The response is: "Sure! [Retrieval] Alpacas and llamas are both domesticated species of South American camelids. [Continue to Use Evidence] Alpacas are much smaller than llamas, with a shoulder height of 3 to 4 feet. [Continue to Use Evidence] They are also bred specifically for their fiber, which is used to make all sorts of textiles and clothing." This indicates that the instruction requires additional retrieval to answer and continues to utilize the retrieved evidence.

As shown in Table 3, our ablation studies reveal significant insights about the model's architectural dependencies. The removal of the Critic module, which serves as the quality control mechanism for generated outputs, leads to a measurable performance drop of approximately 3.2 points on our evaluation metric. This degradation is particularly evident in output coherence and factual consistency. More dramatically, when we disable the retrieval module - the core component that fetches relevant information from external knowledge sources - we observe a sharp performance decline across all metrics, with some tasks showing up to 40% degradation in output quality [15]. The most severe impact comes from removing accuracy-related constraints or optimizations, which causes a substantial 16.9-point drop in factual accuracy, highlighting how critical these safeguards are for maintaining the model's reliability. These results collectively demonstrate that both the retrieval functionality and accuracy-preserving mechanisms are indispensable components of our architecture. Interestingly, in contrast to these critical modules, removing the "Knowledge Coverage" (KC) module - designed to optimize memory usage - shows only a minor impact on performance (less than 1 point difference), suggesting it plays a more auxiliary role in our system.

Table 4 presents our comprehensive benchmarking results across multiple model architectures. The baseline Llama2-7B model achieves 38.8% accuracy on our carefully designed evaluation set, which consists of 5,000 domain-specific questions requiring both factual knowledge and reasoning

capabilities. Through instruction fine-tuning, the Alpaca-7B variant shows significant improvement, reaching 48.2% accuracy. The more advanced Mistral-7B architecture, with its innovative attention mechanisms, further pushes performance to 51.3%, demonstrating the value of architectural advancements. Our-RAG-7B model, which integrates real-time retrieval capabilities with the language model's generative power, achieves a new state-of-the-art accuracy of 54.8% - representing a 16% improvement over the original Llama2-7B baseline and outperforming all other compared models by at least 3.5 percentage points. These results not only demonstrate the effectiveness of the retrieval mechanism in enhancing language model capabilities but also suggest that the combination of retrieval and generation creates synergistic benefits that exceed what either approach can achieve independently.

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